

Deconfounded Human Skeleton-based Action Recognition via Causal Intervention

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ABSTRACT

Human action recognition has vast research prospects in the field of virtual reality, particularly in the context of the skeleton modality with enhanced robustness to background noise. However, most recent researches overlook a dataset bias caused by the confounder *subject*, namely the customized and *subject*-specific sub-actions correlated spurious causalities between actions and predicted labels. In this work, we build a skeleton causal graph to formulate the causalities and causally intervened the confounded recognition model via backdoor adjustment, and design a novel causal intervention based pipeline CI-GCN for skeleton-based action recognition to deconfound model and improve recognition accuracy, which achieves the state-of-the-art on three public action datasets.

Index Terms: Action Recognition, Human Skeleton, Graph Convolution, Causal Intervention, Confounder.

1 INTRODUCTION

With the surge in real applications scenarios of human behavior analysis and understanding, e.g., anomaly detection, virtual reality and human-computer interaction, classic action recognition tasks have become the focus of research in VR and CV communities. Action recognition is aimed at empowering machines to mine latent motion patterns from different data modalities (RGB videos, depth images, point clouds and human skeletons) then establishing the accurate mapping to action labels.

However, recent methods have failed to take into account the training disturbance caused by a dataset bias from individualized *subject*-specific actions. In Fig.1, different *subjects* performed the same category of action ("taking a selfie"), namely *lifting the phones—turning towards the camera—pressing the shutter*. This process usually generates category-standard dependencies among certain highly semantically related joints like the head, neck, wrist, thumb and elbow. But the individuality of different *subjects* may lead to variability in movements (*subject 2* added a unique gesture, *subject 5* held a bag and *subject 9* was seated on a chair). The *subject*-specific joints dependencies are exactly triggered by these *subject*'s own customs. Assuming that the action samples from *subject 1-6* are utilized for training, these dependencies may cause an inference bias, that is, misclassifying the action "taking a selfie" performed by *subject 9* as "pointing".

Our initial focus should be on "Who caused it and why it happened?". We believe that the root of this issue lies in *subject* variation, wherein *subjects* express the same action semantics with

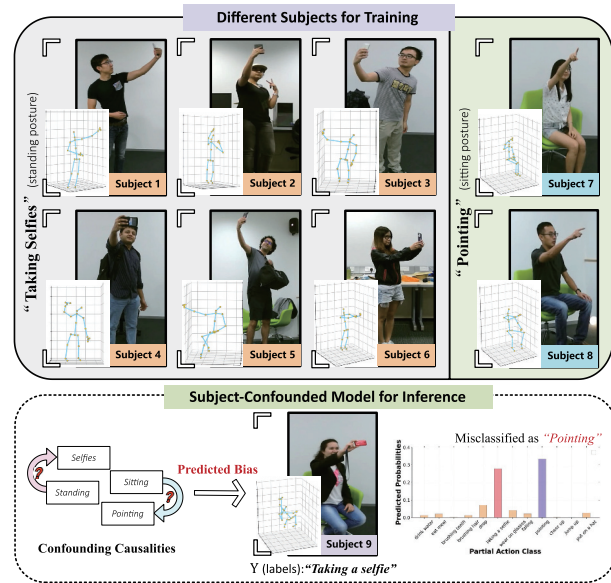


Figure 1: The schematic diagram of confounder *subject*.

their unique poses. Concretely, let X represent the input skeleton-based action tensor and Y represent the corresponding category prediction. Our goal is to design a network that strives to approximate a conditional probability $P(Y|X)$. We find that the *subject* S is a confounder for $P(Y|X)$ due to inject the spurious correlation $X \leftarrow S \rightarrow Y$, so next we respond to "How to solve it". We pursue substituting $P(Y|X)$ with a new training objective $P(Y|do(X))$, where the *do*-operation aims at modeling the causality between *cause* X and *effect* Y in the absence of confounding effort. In this work, we build a skeleton causal graph (SCG) in Fig.2 (b) to formulate the causalities among skeleton actions, joint dependencies, *subjects* and predicted labels, and further develop a novel skeleton causal intervention module (SCI), employing the backdoor adjustment (cut off the backdoor path, e.g., $X \leftarrow S \rightarrow Y$) to deconfound *subject* for approximating $P(Y|do(X))$.

2 METHOD

We formulate the causalities among skeleton actions X , *subjects* S , joint dependencies D and predicted labels Y with a skeleton causal graph (SCG).

Skeleton Causal Graph. 1) $S \rightarrow X$. *Subject* S performs a certain action, which is converted into skeleton mode X by motion capture. S determines the movement of each joint in X . **2) $X \rightarrow D \leftarrow S$.** There are different dependencies D among human joints when an action is performed. The semantics of the skeleton action X determines which joints have strong dependencies and which ones are weak. Additionally, D can also be classified as both category-standard and *subject*-specific dependencies. **3) $X \rightarrow Y \leftarrow D$.** In

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Table 1: Comparison of recognition accuracy with state-of-the-arts on NTU-RGBD 60, NTU-RGBD 120 and Northwestern-UCLA datasets. J_s and B_s mean the joint stream and the bone stream. * indicates the accuracy reproduced with officially released code.

Year	Methods	NTU-RGBD 60						NTU-RGBD 120						NW-UCLA	
		X-Sub(%)			X-View(%)			X-Set(%)			X-Sub(%)			Top-1(%)	
		J_s	B_s	$J_s + B_s$	J_s	B_s	$J_s + B_s$	J_s	B_s	$J_s + B_s$	J_s	B_s	$J_s + B_s$	$J_s + B_s$	
TVCG'23	LKA-GCN[3]	89.0	—	90.7	95.0	—	96.1	84.0	—	87.8	82.6	—	86.3	—	
CVPR'21	CTR-GCN*[1]	89.6	90.2	92.0	94.9	94.8	96.3	86.4	87.3	90.0	84.9	85.7	88.7	95.5	
	Ours	90.9	91.4	92.7	95.7	95.6	96.7	87.8	88.5	90.8	86.2	87.4	89.5	96.4	
ICCV'23	HD-GCN*[2]	90.4	90.7	92.4	95.3	95.0	96.4	87.1	88.1	90.3	85.4	86.3	89.0	96.6	
	Ours	91.3	91.6	92.8	95.9	95.5	96.8	88.4	89.3	91.2	86.5	87.7	90.0	96.9	

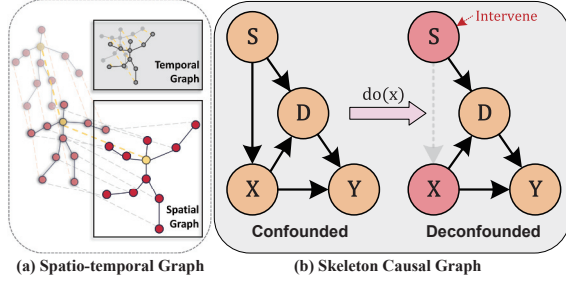


Figure 2: (a) Spatio-temporal Graph. (b) Skeleton Causal Graph.

general, S does not explicitly determine the predicted label Y , meaning there is no direct causal edge $S \rightarrow Y$. Therefore, the causal path needs to pass through the mediate vertex X or D . Clearly, there is a causal correlation from X to Y , linked by a recognition model $P(Y|X)$. Moreover, joint dependency D also affects Y , namely $D \rightarrow Y$. On this basis, S becomes a crucial vertex in causal graph. However, we can find that a backdoor path $X \leftarrow S \rightarrow D \rightarrow Y$ originating from S confounds X and Y . Even when $X \rightarrow Y$, the *subject* variation may still establish a spurious correlation from X to Y .

Causal Intervention via Backdoor Adjustment. We have encountered a question worth pondering: how to estimate $P(Y|do(X))$? We leverage backdoor adjustment to approximate it, that is, cut off the backdoor path by removing the causal edge $S \rightarrow X$ in the SCG. Therefore, $P(Y|do(X))$ is defined as:

$$P(Y|do(X)) = \sum_{i=1}^{\eta} P(Y|X, D = \phi(X, s_i)) P(s_i), \quad (1)$$

due to $S \rightarrow X$, it causes $P(s|X) = P(s)$, so that a fair summation without confounder s_i is performed. $P(Y|do(X))$ learns the average causal effects for each category caused by s_i from observed data.

Skeleton Causal Intervention. We design a skeleton causal intervention module based on the above backdoor adjustment. Specifically, to implement Eq.1, we refer to two tricks, the first one: *Normalized Weighted Geometric Mean* (NWGM) can approximate the expectation to feature-level, namely move the outer $\sum_{i=1}^{\eta}$ inside. Therefore, Eq.1 is transformed into:

$$\begin{aligned} P(Y|do(X)) &\approx P\left(Y|X, \bar{D} = \sum_{i=1}^{\eta} \phi(X, s_i) P(s_i)\right) \\ &= \text{SOFT}(f_Y(X, \bar{D} = E_s[\phi(X, s)])), \end{aligned} \quad (2)$$

where $f_Y(\cdot)$ calculates the logits with M action categories and $\text{SOFT}(\cdot)$ denotes the softmax layer. $\bar{D}$ is the causally intervened joint dependency. E_s represents the expectation for the *subject* s . The second trick: $f_Y(X, \bar{D})$ can be approximated by a linear model:

$$f_Y(X, \bar{D}) = \mathbf{W}_X X^{(L)} + \mathbf{W}_{\bar{D}} \bar{D} \quad (3)$$

where $X^{(L)} \in \mathbb{R}^d$ denotes the action representation containing the high-level semantics extracted from $X^{(0)} \in \mathbb{R}^d$ by the SFE. d is a

simplified notation for the dimension of the skeleton feature. And $\mathbf{W}_X, \mathbf{W}_{\bar{D}} \in \mathbb{R}^{M \times d}$ are the learnable parameters of two fully connection layers, respectively. M represents the number of action categories. $X_{\bar{D}}$ can be regarded as an approximation of $\bar{D}$.

$X_{\bar{D}}$ assumes that we deconfounded the *subject* s , but it is difficult to collect all *subjects*. Therefore, we construct a confounder dictionary $S = \{s_1, \dots, s_M\} \in \mathbb{R}^{M \times d}$ from training dataset, including M action categories. s_i denotes the average skeleton feature of the i -th action category generated by the pre-trained backbone, called action prototype. Specifically, we maintain M skeleton feature buckets $\{\phi(X_q^i)\}_{q=1}^{Q_i}$, $i \in [1, M]$ corresponding to M categories, where Q_i represents the number of action samples in the i -th category and ϕ is the backbone. The formula of the action prototype is $s_i = \frac{1}{Q_i} \sum_{q=1}^{Q_i} \phi(X_q^i)$, which contains the customized joint dependencies from different *subjects* for the i -th action category. In this way, $X_{\bar{D}}$ can be calculated as the weighted summation of s_i .

3 EXPERIMENTS

Our CI-GCN exhibits competitive performance with most state-of-the-arts, which achieves a significant improvement in recognition accuracy over CTR-GCN under each benchmark on three datasets, such as 1.3% with J_s and 1.2% with B_s on NTU60-XSub, and 1.3% with J_s and 1.7% with B_s on NTU120-XSub. Interestingly, by replacing the backbone with HD-GCN, Tab.1 has proven that the CI-GCN pipeline can actually achieve further performance improvements (+1.3% and +1.1% under X-Set and X-Sub with joint stream on NTU120). Potentially, we hope this pipeline will enable SOTAs to pursue a purer causality, thereby improving the accuracy.

4 CONCLUSION

The primary objective of this work is to improve the performance of *subject*-confounded recognition models by focusing on the pursuit of true causal correlations. The causalities within skeleton-based action recognition are modeled by the skeleton causal graph (SCG). By employing causal inference, we intervene on the confounder *subject*, effectively blocking backdoor paths. A novel causal inference based recognition pipeline CI-GCN was integrated, and we verified its effectiveness and compatibility on three public datasets. In future work, CI-GCN will continue to be generalized to other action modalities, and explore the detailed VR applications.

REFERENCES

- [1] Y. Chen, Z. Zhang, C. Yuan, B. Li, Y. Deng, and W. Hu. Channel-wise topology refinement graph convolution for skeleton-based action recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 13359–13368, 2021. 2
- [2] J. Lee, M. Lee, D. Lee, and S. Lee. Hierarchically decomposed graph convolutional networks for skeleton-based action recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 10444–10453, October 2023. 2
- [3] Y. Liu, H. Zhang, Y. Li, K. He, and D. Xu. Skeleton-based human action recognition via large-kernel attention graph convolutional network. *IEEE Transactions on Visualization and Computer Graphics*, 29(5):2575–2585, 2023. doi: 10.1109/TVCG.2023.3247075 2